

■ HACKATHON PROJECT · WEB DEVELOPMENT ■

Likho Write. Publish. Share.

A lightweight blogging platform built with pure HTML, CSS and JS –
designed for first-time writers.

PRESENTED BY TEAM 4 : UEM NEWTOWN(IEM KOLKATA)

RUNTIME TERRORS

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CREATION FRICTION

Why is publishing a blog so overwhelming for beginners?



01

1. Hosting is needed

Vercel, Netlify, or something similar. Create an account, deploy it.

• DEPLOYMENT STEP

ACCOUNT • DEPLOY



02

2. Or heavy software

WordPress needs a database. Publii is a 100 MB+ Electron app.

• HEAVY SETUP

100 MB+ APP



03

3. Static site generator?

Hugo builds it, but you still have to host it yourself.

• STILL HOSTED

SELF-HOSTING

THE OFFLINE PROBLEM

And an internet connection is definitely needed.

If you want to write offline, or show it to someone-it's difficult.

How it works?



01

1. Double-click on run.bat

The batch file just runs `likho.exe`. Nothing else.

```
run.bat -> likho.exe
```



02

2. Small server starts inside

Written in Go, HTML, CSS, and JS are all embedded inside the binary.

```
Go binary + embedded assets
```



03

3. Browser opens automatically

`localhost:4000`

```
localhost:4000
```



04

4. Local Markdown Storage

Whatever you write gets saved into a file `blog-data/posts/*.md` – You can open it with Notepad too.

```
blog-data/posts/*.md
```

FEATURE HIGHLIGHT

The most fun part

The server listens on the entire network, not just localhost.



Press the QR button in the header.



Scan it with your phone.



The blog opens on your phone.



Anyone on the same Wi-Fi can read it.



Instant Local Share

Scan & Read instantly

 No hosting

 No domain

 No internet

 No money

Connecting ideas

<> Type [[name of another post]] -> becomes an instant link



Post doesn't exist?

AUTO-CREATE

Click the link, and the editor will open with that name. Create it right there.



Interactive Map Page

GRAPH VIEW

Shows the whole web - which idea is connected to which. You can move the nodes around.



Automatic Backlinks

BACKLINKS

Below every post, it shows where all incoming links originated from.



LIVE KNOWLEDGE WEB

5 MODES ACTIVE



Interactive Graph Map

Click node to inspect



Inspired by Obsidian - brought into a live blog, not a static site generator.

LIVE GRAPH ARCHITECTURE

Where people get bored

100% PRIVATE & LOCAL

^v Medium tells you what % of people read the post.

Likho tells you which paragraph is driving them away.



Dark, vibrant strip

That's how far readers actually read through.

HIGH ATTENTION



Where color fades

That exact paragraph is where people run away.

DROP-OFF POINT

BUILDING SOFTWARE THAT LASTS

LIVE READER HEATMAP

PARAGRAPH 1 - 100% READ

48 READERS

Most modern web frameworks demand constant updates, complex deployment pipelines, and heavy cloud infrastructure just to display text.

PARAGRAPH 2 - 65% READ

35 READERS

When you strip away external dependencies and store everything in local files, speed increases by an order of magnitude.

PARAGRAPH 3 - 12% READ (FADED)

6 READERS

However, configuring custom build tools and webpack plugins for static site generation can easily become frustrating...

Reader drop-off detected here



No cookies. No tracking ID. No Google Analytics. Just a counter, in the same text file, on this computer.

What's new, and what isn't

ALL FEATURES

STANDARD (7)

INNOVATIONS (5)

These are normal things

EXPECTED BASELINE

 Search	Baseline	 Dark mode	Baseline
 Markdown editor	Baseline	 Tags	Baseline
 Revision history	Baseline	 Related posts (TF-IDF)	Baseline
 Reading time	Baseline		

This is different

LIKHQ ARCHITECTURE

 The blog itself is an .exe file	Standalone Binary
 The entire room can read via QR	Instant Network Share
 Works without internet	100% Offline
 Paragraph-level drop-off	Paragraph Analytics
 Carry the whole blog in a single file	Ultra Portable

 We aren't claiming to have invented 12 different things. We chose an architecture, and these are its benefits.

ARCHITECTURAL CHOICE

How it was built

Built for extreme portability, zero external runtime, and minimal footprint.



Go, a single static binary

CGO_ENABLED=0 – no runtime needed. 9.7 MB on Windows executable.

9.7 MB



Only 2 dependencies

goldmark (markdown) and go-qrcode. Everything else is standard library.

2 DEPS



No database

Markdown files + a small front-matter parser.

FLAT FILES



No frontend framework

Vanilla JS. The graph canvas uses a custom, hand-written force simulation.

0 FRAMEWORKS



Built on Linux, for Windows

Cross-compiled in Docker, then tested in Wine.

CROSS-COMPILED

ZERO DEPENDENCY FOOTPRINT

Standalone single-executable deployment with zero cloud server overhead.

Likho Binary Inspection

BUILD & ARCHITECTURE MATRIX



COMPILER

Go, a single static binary

CGO_ENABLED=0 – no runtime needed. 9.7 MB on Windows executable.

```
$ CGO_ENABLED=0 GOOS=windows GOARCH=amd64 go build -ldflags="-s -w"
  • Compiling likho.exe (pure static Go executable)
  • Output Binary Size: 9.7 MB (Zero C dependencies)
```

Click any architecture block to inspect technical specs and build parameters.

STEP 1 / 5

Go 1.22 • CGO_ENABLED=0 • Wine Approved

What's not there yet & What's next

Honest constraints of a zero-cloud binary and what we're building next.

⚠️ What's not there yet

KNOWN CONSTRAINTS



No login

Anyone on the same Wi-Fi can edit. Fine for a home/college network, but not for cafe Wi-Fi.

LAN OPEN ACCESS



No image upload

External URLs work.

URL ONLY



Only one person should edit

If two people edit at the same time, the last one wins.

LAST WRITE WINS

NO CLOUD DEPENDENCY Likha remains 100% local, lightweight, and zero-configuration.

⚡ What's next

UPCOMING FEATURES



Dynamic connection lines

Thicken or thin the map's connection lines based on which link the reader clicks.

IN DEVELOPMENT



Live LAN reader presence

"3 people are currently reading" live on the LAN.

PROTOTYPED

● LIVE PREVIEW: LAN READER COUNTER

LIVE ON LOCAL NETWORK



3 people are currently reading

192.168.1.x



Click nodes to simulate reader link activity

CLICK-WEIGHTED MAP PREVIEW

LAN Mode • Zero Server Setup • Likha Core

> QUICKSTART & LIVE DEMO

Try running it yourself

01

Extract zip

Unzip the archive anywhere on your machine.

02

Double-click run.bat

Launches standalone likho.exe instantly.

03

Done!

Opens automatically in browser at localhost:4000.

● LIVE WI-FI ACCESS

Or scan the QR, it will open right now in this room.



⊞ http://192.168.1.42:4000



Likho · github.com/likho-blog/likho

🌟 Star on GitHub

WHAT'S NEXT

Thank you. Likho is just getting started.

User login and authentication.
Comments on every post.
Likes and engagement counts.
Categories and tags.
AI-assisted blog suggestions.
Dark mode theme.

